

Research Portfolio

Social Network Analysis of Peer Support and Group Formation

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Project Title

Peer Support, Individual Popularity, and Group Formation in MRAD Course

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Project Overview

This project examined how peer support and interaction patterns relate to individual popularity and group formation among master's students in psychology. The study focused on how academic and non-academic interactions between students could be represented as social networks and analysed using social network indicators.

The project was developed as part of my Master's training in Social Psychology and strengthened my interest in understanding how individual-level social behaviour can develop into broader group-level structures.

Research Focus

The project explored several questions related to peer support and social interaction, including:

- How discussion frequency relates to students' degree centrality within a peer network
- How different forms of peer support contribute to individual network position
- How academic and non-academic interaction networks differ in centrality and clique formation
- How group formation can be understood through social network structures

Data and Participants

The study involved interaction and survey data from psychology master's students at Universitas Indonesia. The data included:

- Survey responses on peer support and discussion frequency
- Interaction diary data on academic and non-academic student interactions
- Network data representing who interacted with whom in academic and non-academic contexts

The original dataset is not shared publicly to protect participant privacy.

Methods and Tools

This project used social network analysis to transform student interaction data into measurable network structures.

Methods used:

- Social network analysis
- Directed graph construction
- Degree centrality analysis
- Normalized degree centrality analysis
- In-strength and out-strength analysis
- Clique identification
- Academic and non-academic network comparison
- Data cleaning and network visualization

Tools used:

- R
- RStudio
- `igraph`
- Data visualization in R

My Contribution

My contribution included preparing and analysing the interaction network data, translating social-psychological constructs into measurable network indicators, and interpreting the results in relation to peer support and group dynamics.

Key contributions:

- Cleaned and prepared academic and non-academic edgelist data
- Constructed graph objects from interaction data using R and `igraph`
- Calculated centrality indicators to identify students' network positions
- Analysed clique formation to understand subgroup patterns
- Compared academic and non-academic interaction networks
- Connected network indicators with social-psychological concepts such as peer support, popularity, and group formation
- Prepared visual and analytical outputs for research presentation

Key Learning

This project helped me understand how abstract social-psychological phenomena can be operationalised into observable and quantifiable behavioural patterns. It trained me to think not only about what people report, but also about how their interactions form structures, positions, and subgroups within a social system.

Although this project focused on student interaction networks rather than physical crowd movement, it provided a strong foundation for studying how individual behaviours aggregate into group-level patterns.

The project also shows my readiness to work with interdisciplinary methods, including behavioural observation, quantitative analysis, and computational approaches to social behaviour.

Ethical Note

The original data are not included in this portfolio due to privacy and confidentiality considerations. Any technical demonstration shared publicly will use anonymized or synthetic data only.